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Higher Cardiovascular Risk Factor Burden Associated With Cumulative Concussion History and Age-Related Differences

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Objective: Growing evidence suggests cardiovascular dysfunction represents 1 pathway by which recurrent concussion may lead to adverse long-term outcomes, with earlier onset increasing risk. We investigated the interaction of concussion history and age on the prevalence of cardiovascular risk factors (CRF) and the association of CRF with multidomain function across age in former professional football players. **Setting:** Survey-data **Participants:** Former National Football League players. **Design:** Former players were dichotomized into older (n = 1071; > age 50-years of age) and younger (n = 902; ≤ 50 years of age) age groups. Logistic and linear regression were used to test the interaction of concussion history and age on prevalence of CRFs, and the association of cardiovascular risk factors with functional outcomes across age groups. **Measures:** CRFs included the following: hypertension, hyperlipidemia, diabetes, and sleep apnea. Clinical outcomes from the Patient-Reported Outcomes Measurement Information System included Depression, Anxiety, Cognitive Function, and Emotional and Behavioral Dyscontrol. Logistic and linear regression tested the interaction of concussion history and age on prevalence of CRFs, and the association of cardiovascular risk factors with functional outcomes across age groups. **Results:** Significant concussion history by age interactions were observed for hypertension and hyperlipidemia, (P's < .001), with greater number of prior concussions being associated with higher rates of diagnoses for younger former athletes only. A greater prevalence of sleep apnea with higher concussion history was observed regardless of age (P < .001). Significant age by hypertension and hyperlipidemia interactions were observed on clinical outcomes, with higher symptom endorsement among those with CRF in younger former players (P's < .001). **Conclusion:** Given higher rates of diagnoses at earlier ages, findings support cardiovascular health as representing one pathway by which recurrent concussion may lead to adverse long-term outcomes in former contact sport athletes. A clinical preventative approach including early screening and management of CRF in those with greater history of concussion should be considered. **Key words:** *brain injury, cardiovascular risk, concussion, diabetes, hyperlipidemia, hypertension, sleep apnea, TBI*

INTRODUCTION

There is growing concern surrounding adverse long-term neurological health among those who sustain repetitive neurotrauma, such as cumulative concussion and repetitive head impacts that do not result in concussive injury during sport. While some studies have reported increased risk of neurodegenerative disease in

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former contact sport athletes later in life, not all studies have observed this association, and findings have been highly inconsistent.[1–5] One potential source of this variability across studies involves the fact that there are multiple pathways by which repetitive neurotrauma may increase adverse outcomes later in life, and the degree of risk depends on the progression of different processes.[6] These pathways may occur directly (eg, white matter injury/susceptibility, ongoing neuroimmune signaling alterations) or indirectly (eg, chronic sleep disturbance associated with cumulative concussion) as a consequence of repetitive neurotrauma.[7-9] The collective consideration of these different pathways is likely the most optimal approach to truly understanding the association between repetitive neurotrauma and adverse neurological health later in life.

Cardiovascular health is a well-recognized risk factor for Alzheimer's disease and related dementias (ADRD) and represents a leading potential indirect pathway by which neurotrauma may increase risk for adverse neurological outcomes later in life.[10] Head injury is associated with greater odds of cardiovascular disease (CVD), with higher rates observed in those with greater injury severity.[11,12] Select prospective studies have also demonstrated that individuals who sustain a TBI are at increased risk for developing incident CVD and CVD risk factors longitudinally over time, and the risk is dose-dependent based on injury severity.[13] Depending on a number of factors (eg, level of play, division), higher rates of cardiovascular risk factors (eg, hypertension/prehypertension) and indicators of poor cardiovascular health (eg, higher systolic blood pressure, lower high-density lipoprotein) have been observed in former contact sport athletes.[14-16] The role of repetitive neurotrauma as a potential driver of higher rates of CVD risk factors in contact sport athletes was supported by a study of former professional football players that showed higher rates of hypertension, hyperlipidemia, and overall CVD risk factor burden were associated with greater concussion symptom burden during play.[17]

High CVD risk factor burden before the age of 50 is associated with increased mortality risk by at least 10 years and a higher risk of CVD/related events.[18,19] Earlier onset of CVD (before the age of 60) confers greater risk for adverse neurological outcomes later in life, and prior studies have shown that CVD onset in midlife is more strongly associated with dementia risk.[20] As such, it is necessary to investigate whether the association between repetitive neurotrauma and CVD risk factors differs earlier and later in the lifespan. In other words, does repetitive neurotrauma increase the risk for CVD risk factors earlier in life when they are less likely to be present due to age and aging-related processes? Repetitive neurotrauma (ie, a greater number of prior concussions) is associated with worse ratings of clinical outcomes across a number of domains in former athletes.[21-23] However, given the potential influence of repetitive neurotrauma on CVD risk factors, it is necessary to better understand the contribution of CVD risk factors on health and function (eg, psychiatric and neurobehavioral), particularly earlier in the lifespan. As such, the objectives of the current study were to (1) determine the influence of prior concussions and interactions with age (younger compared to older [> 50 years of age]) on the prevalence of CVD risk factors in former professional football players; and (2) investigate the relative influence of CVD risk factors on clinical outcomes (depression, anxiety, subjective cognition, and neurobehavioral symptoms) in younger compared to older former players.

METHODS

Participants

Former professional American football players with at least 1 year of play in the National Football League (NFL) were enrolled through multiple means, including individual team and organizational (eg, NFL Alumni Association) communications (eg, newsletter notification, mailed letters) via email and hard copy questionnaires. Following initial recruitment, the response rate was estimated to be approximately 12%. From that

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point, ongoing recruitment and rolling enrollment occurred through peer referral. At the time of the current analyses, 1999 former players completed a general health survey (GHS; see Brett et al and Walton et al for details on development and validation)[21,22,24]. Of these respondents, data were missing for body mass index (BMI) (n = 16), race (n = 9), and age (n = 1), resulting in a final sample of N = 1973. The study was approved by the University of North Carolina at Chapel Hill and Medical College of Wisconsin Institutional Review Boards, and participants provided informed consent. Due to the sensitive nature of the data, support data are not available, though reasonable requests may be submitted to the corresponding author (B.L.B.).

Concussion history, cardiovascular risk factors, and function

The GHS included demographic information, health, sport, and concussion history, as well as various patient-reported outcomes across a range of domains. Participants reported the number of prior concussions (diagnosed and undiagnosed) based on a standard definition of concussion that was presented: "a blow to the head that is followed by a variety of symptoms that may include any of the following: headache, dizziness, loss of balance, blurred vision, "seeing stars," feeling in a fog, or slowed down, memory problems, poor concentration, nausea, or throwing up. Getting "knocked out" or being unconscious does NOT always occur with a concussion."[25] Sport-related concussions at all levels of football play and nonsport concussions were included in the final count. Consistent with prior studies in this cohort, concussion history was categorized into 5 groups: 0, 1-2, 3-5, 6-9, and 10 + .[26]

CVD risk factors for consideration included those reported in prior studies examining the influence of concussion history in former professional football players, such as hypertension, hyperlipidemia, and diabetes.[17,27] Additionally, given the higher rates of obstructive sleep apnea (OSA) in former professional football players, reported associations of OSA with tau and clinical function in former players, and the established influence of OSA as a CVD-related dementia risk factor in the general population, OSA was also included in the study.[28,29] Participants reported the presence/absence of each diagnosis based on queries of "have you ever been told by a clinician that you have the following diagnosis." In addition to individual diagnoses, cumulative CVD risk factor burden was calculated from the sum of the diagnoses endorsed and was collapsed into groups 0, 1, and 2+ based on the sample distribution.

Previously published findings from an earlier subset of this cohort demonstrated significant associations between greater concussion history and higher levels of depression, anxiety, cognitive, and neurobehavioral symptoms.[21] As such, primary measures for the current study included the Patient-Reported Outcomes Measurement Information System (PROMIS) Depression, Anxiety, and Cognitive Function-Short Forms, as well as the Quality of Life in Neurological Disorder (Neuro-QoL) Emotional and Behavioral Dyscontrol Short-Form. Raw scores from each measure were converted to *t*-scores (mean =50 and standard deviation [SD] = 10) based on normative data obtained from the general population. Higher scores on the Depression, Anxiety, and Emotional and Behavioral Dyscontrol measures indicate worse function, while higher scores on the Cognitive Function form indicate better-rated cognition.

Statistical analysis

Logistic regression models estimating adjusted odds ratios with 95% confidence intervals (CI) were fit for each of the 4 CVD risk factors as dependent variables. Models included age group (older [age > 50 years] and younger players), concussion history, the interaction of age by concussion history, and *a priori* covariates of BMI and race (dichotomized as white and nonwhite based on the distribution of the sample) due to their previously observed associations with CVD risk factor prevalence.[17,30] For models of CVD risk factors in which a significant age group by concussion history interaction was observed, post-hoc logistic regression models were fit that were split by age to examine the association of concussion history on CVD risk factor individually for each group. In order to ensure robustness and better characterize findings across age, the primary models testing age by concussion history interactions were fit, dichotomizing age at 40 and 60 years as well. For analyses involving patient-reported outcomes, multiple imputation involving the Markov chain Monte Carlo method (5 additional iterations of complete datasets generated) was used and reported results were based on pooled data from the multiple imputation datasets. Multivariable linear regression

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models with race included as a covariate were fit to test the association with CVD risk factor by age interactions for psychiatric (anxiety and depression), cognitive, and neurobehavioral symptoms. Unstandardized beta coefficients and standard errors were calculated for independent variables. Statistical significance for all analyses was set at $P < .05$.

RESULTS

The sample of former players (N = 1973) had a mean age of 52.0 ± 16.1 years, with 54% (n = 1071) participants > age 50 years of age and 46% (n = 902) ≤ 50 years of age. There were significant age group differences in race, χ^2 (4) = 83.14, $P < .001$, Cramer's $V = .205$, and BMI, $t = 3.04$, $P = .002$, Cohen's $d = 0.138$. The distribution of total CVD burden and individual risk factors for the full sample and age groups is provided in Table 1.

Concussion history and age interaction on CVD risk factors

The proportion of overall CVD-risk factor burden (0, 1, and 2 + conditions) by concussion history displayed a step-wise increase in the younger, but not older age groups (Figure 1). A significant concussion history by age interaction was observed for hypertension and hyperlipidemia, such that greater history of prior concussion was more strongly associated with higher odds of hypertension and hyperlipidemia, in the younger former athlete group compared to the older athlete group (Figures 2 and 3). Specifically, age and concussion history interactions on greater hypertension odds were observed for those with 10 + and 6-9 prior concussions, $P = .003$–$.020$, and the 3-5 group approached significance, $P = .075$ (Table 2). Similarly, the age by concussion history interaction on greater hyperlipidemia odds was

TABLE 1 *Demographic, sport, and cardiovascular history*

	Full sample N = 1973	50 & Under N = 902	Over 50 N = 1071	Statistic[a]	*P*-value
Age, mean (SD)	52.03 (16.1)	36.9 (7.2)	64.8 (8.6)	NA	NA
Years of football, mean (SD)	17.6 (4.4)	17.4 (4.3)	17.4 (4.4)	−1.76	0.07
Body mass index, mean (SD)	30.9 (4.6)	31.2 (4.7)	30.6 (4.5)	3.04	.002
Cardiovascular risk factor				NA	NA
Hypertension	738 (37.4%)	214 (23.7%)	524 (48.9%)		
Hyperlipidemia	484 (24.5%)	122 (13.5%)	362 (33.8%)		
Diabetes	121 (6.1%)	30 (3.3%)	91 (8.5%)		
Sleep apnea	445 (22.6%)	119 (13.2%)	326 (30.4%)		
Cardiovascular risk burden				235.96	<.001
0	910 (46.1%)	580 (64.3%)	330 (30.8%)		
1	540 (27.4%)	195 (21.6%)	345 (32.2%)		
2 +	523 (26.5%)	127 (14.1%)	396 (37.0%)		
Concussion history				9.97	0.041
0	238 (12.1%)	113 (12.5%)	125 (11.7%)		
1-2	322 (16.3%)	166 (18.4%)	156 (14.6%)		
3-5	474 (24.0%)	214 (23.7%)	260 (24.3%)		
6-9	395 (20.0%)	186 (20.6%)	209 (19.5%)		
10 or more	544 (27.6%)	223 (24.7%)	321 (30.0%)		
Racial identity				83.14	<.001
White	1141 (57.8%)	422 (46.8%)	719 (67.1%)		
Black/African American	723 (36.6%)	402 (44.6%)	321 (30.0%)		
Multiracial	57 (2.9%)	43 (4.8%)	14 (1.3%)		
Other	35 (1.8%)	24 (2.7%)	11 (1.0%)		
Native Hawaiian or other Pacific	12 (0.6%)	9 (1.0%)	3 (0.3%)		
American Indian or Alaskan native	4 (0.2%)	2 (0.2%)	2 (0.2%)		
Asian	1 (0.1%)	0 (0.0%)	1 (0.1%)		
Education				6.75	0.034
Less than a bachelor's degree	294 (14.9%)	122 (13.5%)	172 (16.1%)		
Bachelor's degree	1199 (60.8%)	576 (63.9%)	623 (58.2%)		
Professional degree	478 (24.2%)	203 (22.5%)	275 (25.7%)		

NA = Statistical comparisons of individual cardiovascular disease risk factors performed in full models including interactions and covariates; SD = standard deviation.

[a]Chi-squared test for categorical variables and independent samples *t*-tests for continuous variables; years of football missingness due to participants not reporting participation at all levels or unlikely values (n = 1926).

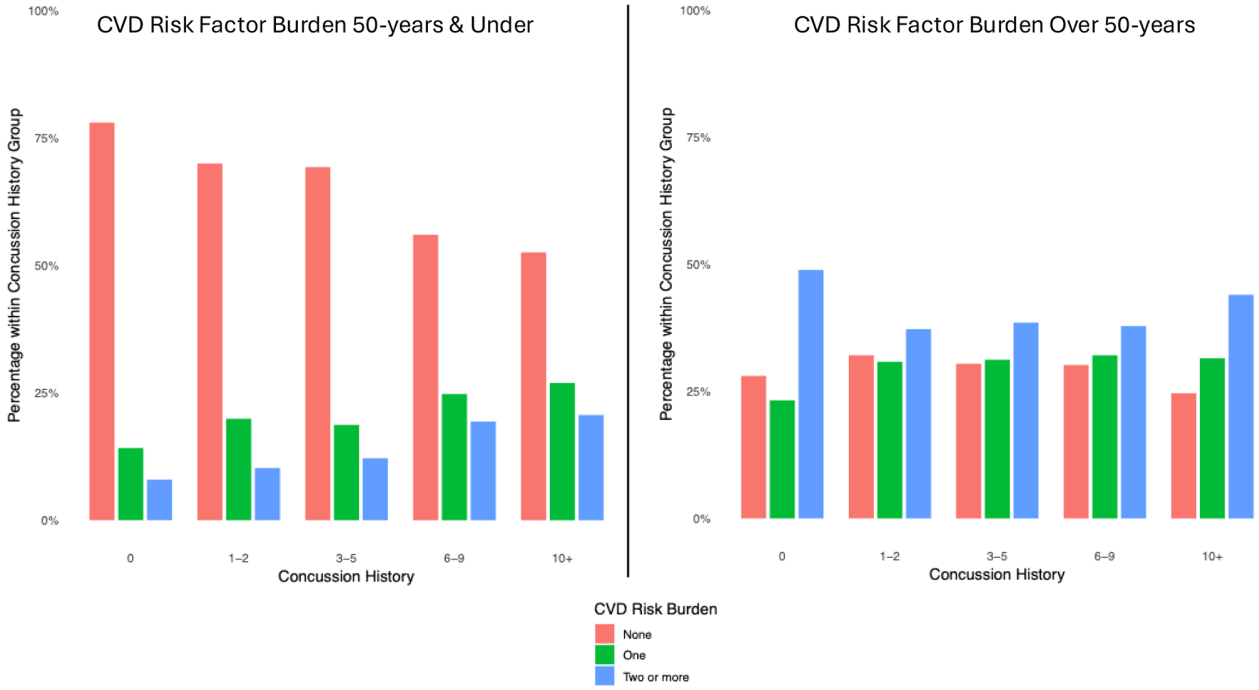

Figure 1. Bar charts showing increasing CVD risk factor burden (0, 1, 2 +) with greater concussion history in the young age group (≤50 years of age), but not older age group (>50 years). Proportion of CVD burden groups reflect 100% of participants within each concussion history group.

observed for those with 10 +, 6-9, 3-5 prior concussions, $P < .001$–$.044$, and a trend for the 1-2 group, $P = .060$.

When split by age group, a general dose-response association between greater history of concussion and hypertension was recorded (ie, higher proportion of individuals with hypertension with increasing number of concussions; Supplemental Digital Content Table 1, available at: http://links.lww.com/JHTR/B23) in the younger group, $\chi^2(4) = 15.70$, $P = .003$, with significantly higher odds of the diagnosis in those with 10 +, (odds ratio [95% CI] = 2.43 [1.35, 4.37], $P < .001$) compared to those reporting no prior head injuries. This trend was not observed in the older age group (Supplemental Digital Content Table 2, available at: http://links.lww.com/JHTR/B23). An overall effect of concussion history and hyperlipidemia was observed, with a similar general dose response relationship also observed in younger former athletes, $\chi^2(4) = 15.34$, $p = .004$ (Supplemental Digital Content Table 1, available at: http://links.lww.com/JHTR/B23). Individual contrasts revealed significantly greater odds of the diagnosis in those with 10 +, OR = 3.27 (1.41, 7.57), $P = .006$, and 6-9 prior concussions, OR = 3.38 (1.44, 7.92), $P = .005$, compared to those with 0 prior concussions. Sensitivity analysis characterizing concussion history groups as an ordinal variable was also significant for hypertension, OR = 1.33 (1.14, 1.55), $P < .001$, and hyperlipidemia, OR = 1.26 (1.12, 1.43), $P < .001$, in younger, but not older, former athletes.

The interaction between age and concussion history group was not statistically significant for diagnosis of OSA, $\chi^2(4) = 5.78$, $P = .216$ (Table 2). However, a significant main effect was observed for concussion history in OSA, $\chi^2(4) = 21.0$, $P < .001$. Specifically, those with 10 +, OR = 3.17 (1.40, 7.15) and 6-9, OR = 2.51 (1.09, 5.79) prior concussions reported greater odds of OSA compared to those with no prior concussions. The interaction between age and concussion history group was not statistically significant, $\chi^2(4) = 6.62$, $P = .158$, for the diagnosis of diabetes. For sensitivity analyses, dichotomizing age at 40 and 60 years, a greater history of concussion at younger ages was associated with higher rates of CVD risk factors (similar to when dichotomized at 50 years), though to a lesser degree when dichotomized at 60 years, as would be expected (Supplemental Digital Content Table 3, available at: http://links.lww.com/JHTR/B23 and Supplemental Digital Content Table 4, available at: http://links.lww.com/JHTR/B23).

Age, CVD risk factors, and clinical function

Regression models testing CVD risk factor by age group interactions were fit to determine the influence of CVD risk factors on patient-reported outcomes across various domains previously shown to be associated with greater concussion history in this cohort. There were significant age

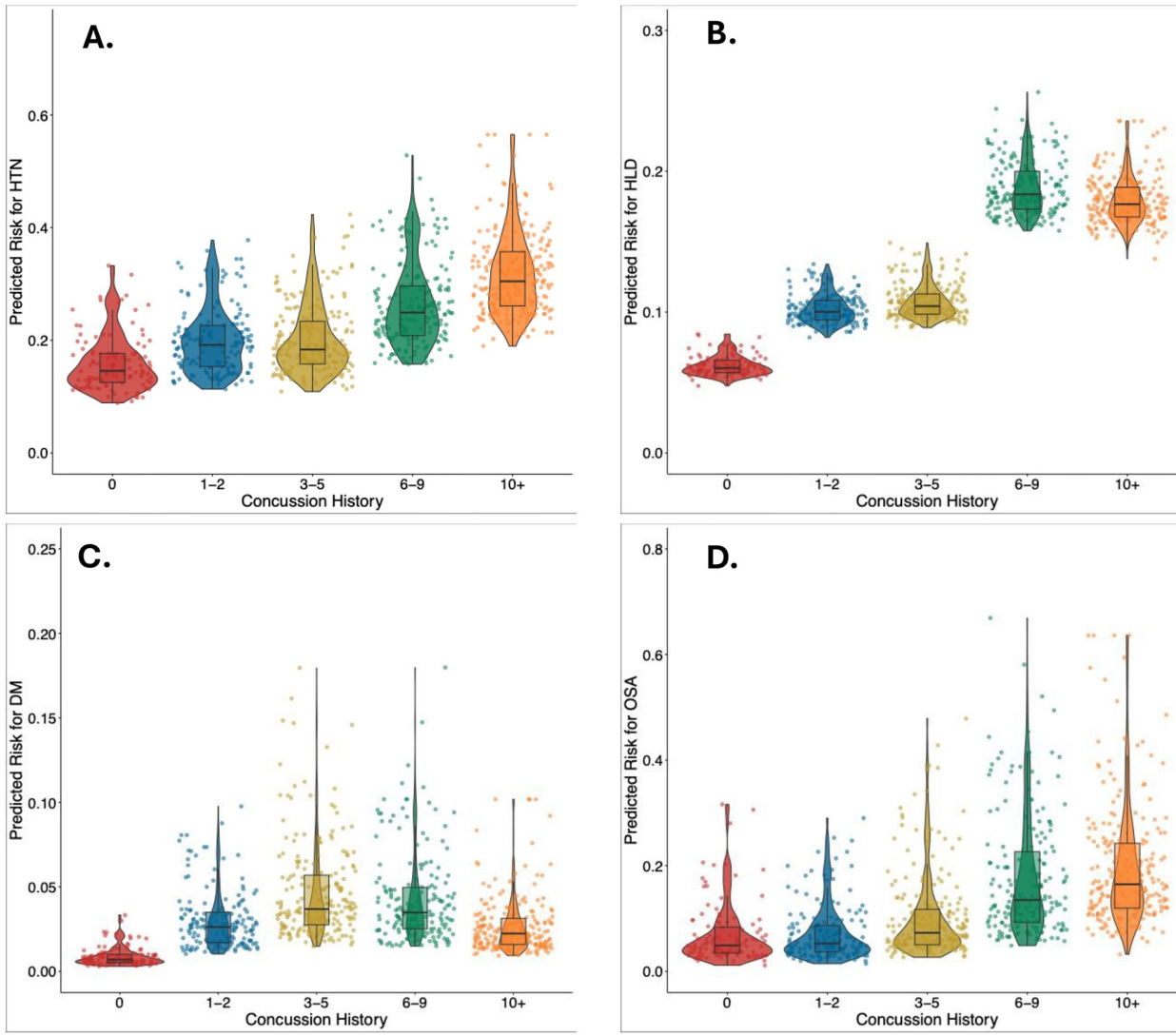

Figure 2. Violin plots of predicted probability of CVD-risk factors by concussion history group in the younger age group (≤ 50-years of age). The predicted odds are based on the full regression models including age, concussion history, age-concussion history interactions, BMI, and race. A general dose-response association between greater history of concussion with hypertension, hyperlipidemia, and sleep apnea was observed.

group by hypertension interactions for all 4 patient-reported outcomes, such that the hypertension was associated with worse functioning across domains for the ≤ 50 years of age group, but not the > 50 years, *P's* < .001 (Table 3). Similar age group by hyperlipidemia interactions were observed for PROMIS Depression, Cognitive Function, and Emotional and Behavioral Dyscontrol, with worse outcomes among those with hyperlipidemia in the younger, but not older age groups (Table 3). Age group by sleep apnea interactions were observed for Emotional and Behavioral Dyscontol, $p = .006$, with worse function in the younger age group reporting a diagnosis of sleep apnea compared to those without, and no differences in the older group (Supplemental Digital Content Table 5, available at: http://links.lww.com/JHTR/B23). Age group by diabetes diagnosis interactions were not observed for any domains of the patient-reported outcomes (Supplemental Digital Content Table 6, available at: http://links.lww.com/JHTR/B23).

DISCUSSION

In this study of former professional football players, a greater history of concussions was associated with a greater prevalence of select cardiovascular risk factors (ie, hypertension and hyperlipidemia) in former athletes ≤50 years of age, but not among those later in the lifespan. Other select cardiovascular risk factors (ie, sleep apnea) were associated with a greater history of concussion regardless of age. Importantly, the presence of these

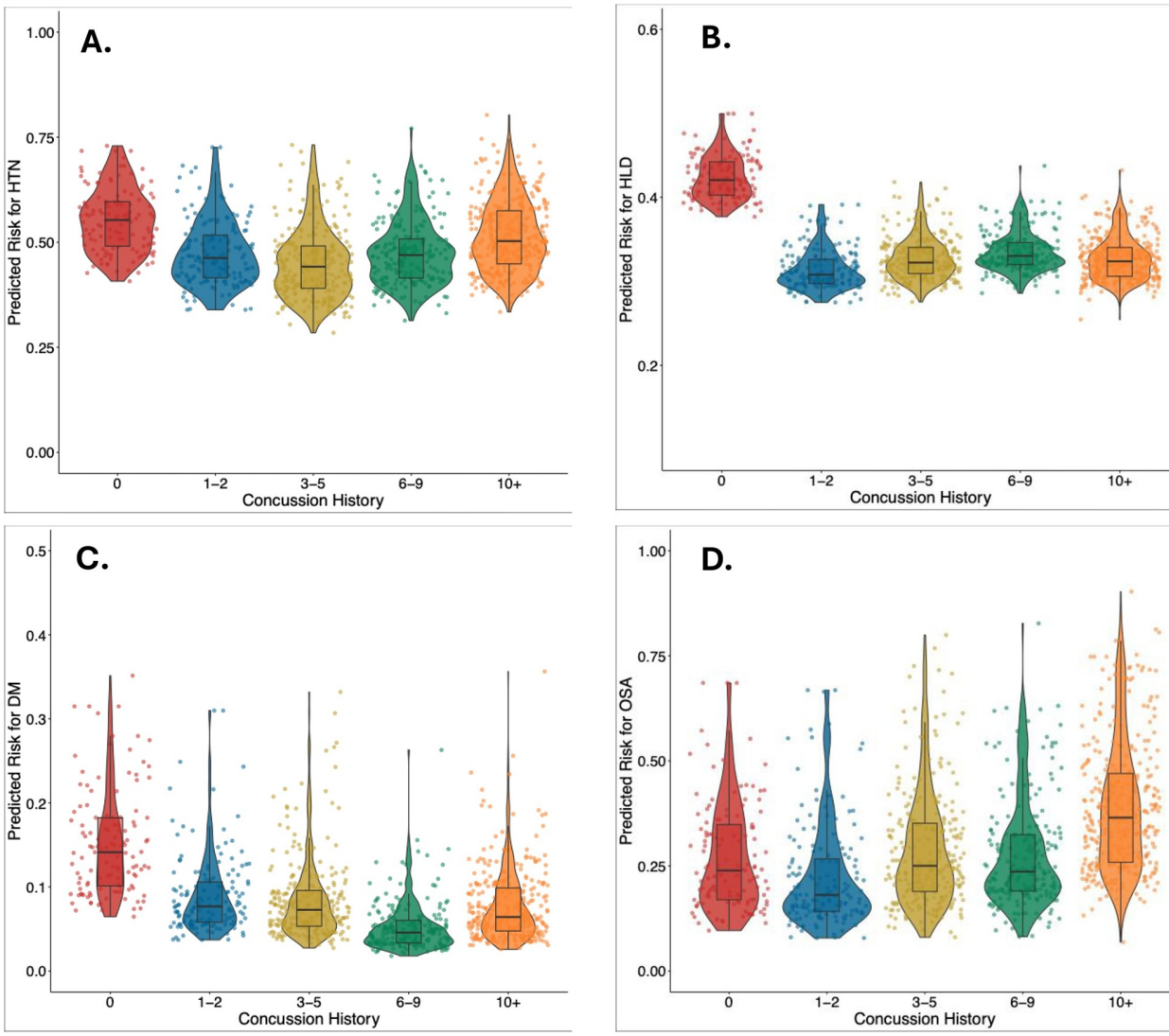

Figure 3. Violin plots of predicted probability of CVD-risk factors by concussion history group in the older age group (> 50 years of age). The predicted odds are based on the full regression models, including age, concussion history, age-concussion history interactions, BMI, and race. An association between greater history of concussion and sleep apnea was observed only in the older age group.

cardiovascular risk factors disproportionally influenced worse functioning across a range of domains, primarily in younger former players, as compared to older players. These findings highlight the importance of early screening and monitoring of cardiovascular risk in former athletes with a greater history of concussion in order to implement earlier intervention, which ultimately has the potential to reduce the risk of cognitive decline and dementia, and improve mental health outcomes.

Greater concussion history: importance of early screening and intervention

The influence of age on the nature of the association between concussion history and cardiovascular risk factors observed in the current study may be a key factor in accounting for discrepancies in the prevalence and influence of cardiovascular risk factors on long-term outcomes in athletes. There are mixed reports on whether cardiovascular risk factors are more prevalent in former athletes compared to the general population.[15,16,31] While sport factors likely play a critical role in explaining these differences (higher BMI in former football players who played offensive line compared to forwards in soccer), the age of the cohort should be taken into consideration based on the current findings.[32] Specifically, odds of CVD risk factors associated with repetitive neurotrauma in former contact sport athletes are more likely to be higher among study samples who are younger in age, as compared to older cohorts, where CVD burden is more prevalent in the

TABLE 2 *Concussion by age interaction effect on CVD risk factor diagnosis*

	Unstandardized coefficient	Standard error	*P*-value	Odds ratio	Lower 95% CI	Upper 95% CI
Hypertension						
Concussion history[a]			.004			
1-2	0.198	0.326	.543	1.22	0.64	2.31
3-5	0.23	0.312	.460	1.26	0.68	2.32
6-9	0.569	0.31	.066	1.77	0.96	3.24
10 or more	0.882	0.298	.003	2.42	1.35	4.33
Body mass index	0.065	0.011	<.001	1.07	1.04	1.09
Age group ψ	1.995	0.318	<.001	7.35	3.94	13.73
Concussion history by age[b]			.032			
1-2 by age	−0.521	0.407	.200	0.59	0.27	1.32
3-5 by age	−0.682	0.382	.075	0.51	0.24	1.07
6-9 by age	−0.894	0.386	.020	0.41	0.19	0.87
10 or more by age	−1.102	0.367	.003	0.33	0.16	0.68
Race^	0.292	0.103	.005	1.34	1.09	1.64
Hyperlipidemia						
Concussion history[a]			.004			
1-2	0.519	0.467	.267	1.68	0.67	4.20
3-5	.584	0.449	.193	1.79	0.74	4.32
6-9	1.226	0.433	.005	3.41	1.46	7.97
10 or more	1.178	0.428	.006	3.25	1.41	7.52
Body mass index	0.025	0.012	.037	1.03	1.00	1.05
Age group ψ	2.422	0.431	<.001	11.27	4.84	26.24
Concussion history by age[b]			.004			
1-2	−0.998	0.53	.060	0.37	0.13	1.04
3-5 by age	−1.011	0.502	.044	0.36	0.14	0.97
6-9 by age	−1.607	0.492	.001	0.20	0.08	0.53
10 or more by age	−1.613	0.48	<.001	0.20	0.08	0.51
Race^	0.011	0.114	.924	1.01	0.81	1.26
Diabetes						
Concussion history[a]			.512			
1-2	1.145	1.105	.300	3.14	0.36	27.39
3-5	1.621	1.058	.126	5.06	0.64	40.22
6-9	1.501	1.07	.161	4.49	0.55	36.57
10 or more	1.079	1.089	.322	2.94	0.35	24.86
Body mass index	0.094	0.019	<.001	1.10	1.06	1.14
Age group ψ	3.199	1.039	.002	24.50	3.20	187.88
Concussion history by age			.158			
1-2 by age	−1.776	1.169	.129	0.17	0.02	1.67
3-5 by age	−2.379	1.111	.032	0.09	0.01	0.82
6-9 by age	−2.713	1.143	.018	0.07	0.07	0.62
10 or more by age	−1.976	1.138	.082	0.14	0.02	1.29
Race^	0.501	0.199	.012	1.65	1.12	2.44
Sleep apnea						
Concussion history[a]			<.001			
1-2	-0.122	0.485	.801	0.89	0.34	2.29
3-5	0.323	0.443	.466	1.38	0.58	3.29
6-9	0.92	0.427	.031	2.51	1.09	5.79
10 or more	1.152	0.416	.006	3.17	1.40	7.15
Body mass index	0.149	0.013	<.001	1.16	1.13	1.19
Age group ψ	1.768	0.433	<.001	5.86	2.51	13.70
Concussion history by age			.216			
1-2 by age	−0.174	0.566	.758	0.84	0.28	2.55
3-5 by age	-0.300	0.511	.557	0.74	0.27	2.02
6-9 by age	−0.908	0.503	.071	0.40	0.15	1.08
10 or more by age	−0.679	0.481	.158	0.51	0.20	1.30
Race^	0.047	0.122	.701	1.05	0.82	1.33

CI, confidence interval.

[a]Reference for concussion history is the zero prior concussion group; ψ = age reference is ≤ 50 years of age group.

[b]Interaction term and odds ratios (<1) display decreasing association of concussion history group level with hypertension and hyperlipidemia in the older compared to the younger age groups (ie, concussion history effect on CVD risk factor is larger in the young group); ^ = racial identity reference is the group identifying as white.

TABLE 3 *Hypertension and hyperlipidemia by age interaction on patient reported outcomes*

	Unstandardized Coefficient	Standard Error	t-statistic 95% CI	p-value 95% CI	Lower	Upper
Hypertension						
Anxiety[c]						
Age group ψ	6.255	0.845	7.40	<.001	4.60	7.91
Hypertension[a]	−1.213	0.611	−1.99	.047	−2.41	−0.02
Age group by hypertension[b]	−3.958	1.007	−3.93	<.001	−5.93	−1.99
Race^	4.691	0.475	9.88	<.001	3.76	5.62
Depression[c]						
Age group ψ	4.522	0.847	5.34	<.001	2.86	6.18
Hypertension[a]	−1.297	0.574	−2.26	.024	−2.42	−0.17
Age group by hypertension[b]	−3.467	0.987	−3.51	<.001	−5.40	−1.53
Race^	3.791	0.461	8.22	<.001	2.89	4.70
Cognitive Function						
Age group ψ	−0.909	0.817	−1.11	.266	−2.51	0.69
Hypertension[a]	1.238	0.610	2.03	.042	0.04	2.43
Age group by hypertension[b]	3.675	0.990	3.71	<.001	1.74	5.61
Race^	−3.120	0.462	−6.76	<.001	−4.03	−2.22
Emo/Behav Dyscontrol[c]						
Age group ψ	6.369	0.901	7.07	<.001	4.60	8.14
Hypertension[a]	−0.333	0.620	−0.54	.591	−1.55	0.88
Age group by hypertension[b]	−5.612	1.075	−5.22	<.001	−7.72	−3.51
Race^	2.427	0.505	4.80	<.001	1.44	3.42
Hyperlipidemia						
Anxiety[c]						
Age group ψ	4.375	1.038	4.21	<.001	2.34	6.41
Hyperlipidemia[a]	0.784	0.646	1.22	.225	−0.48	2.05
Age group by hyperlipidemia[b]	−1.908	1.156	−1.65	.099	−4.17	0.36
Race^	4.913	0.478	10.29	<.001	3.98	5.85
Depression[c]						
Age group ψ	3.638	1.000	3.64	<.001	1.68	5.60
Hyperlipidemia[a]	0.272	0.608	0.45	.655	−0.92	1.46
Age group by hyperlipidemia[b]	−2.524	1.109	−2.28	.023	−4.70	−0.35
Race^	3.988	0.462	8.64	<.001	3.08	4.89
Cognitive Function						
Age group ψ	−1.127	1.055	−1.07	.285	−3.20	0.94
Hyperlipidemia[a]	−0.020	0.647	−0.03	.976	−1.29	1.25
Age group by hyperlipidemia[b]	3.900	1.168	3.34	<.001	1.61	6.19
Race^	−3.298	0.464	−7.11	<.001	−4.21	−2.39
Emo/Behav Dyscontrol[c]						
Age group ψ	5.185	1.110	4.67	<.001	3.01	7.36
Hyperlipidemia[a]	0.703	0.640	1.10	.272	−0.55	1.96
Age group by hyperlipidemia[b]	−3.887	1.238	−3.14	.002	−6.31	−1.46
Race^	2.616	0.508	5.16	<.001	1.62	3.61

CI = confidence interval
[a]Reference for cardiovascular risk factor is the condition absent group.
ψ = age reference is ≤ 50 years of age group.
[b]Interaction term displays decreasing association of hypertension and hyperlipidemia with clinical outcomes in the older compared to the younger age groups (i.e., CVD risk factor has larger effect and is associated with worse function in the young group).
^ = racial identity reference is the group identifying as White.
[c]Robust SE used due to violation of homoskedasticity.

general population.[20,33] Additionally, earlier age of onset for cardiovascular risk factors associated with greater concussion history may be an important factor in the influence of repetitive neurotrauma on dementia risk in former contact sport athletes.

As noted above, cardiovascular risk factors are associated with adverse neurological outcomes later in life, including ADRD, and early intervention to manage these risk factors is important.[10,33] Interventions that address modifiable vascular risk factors, particularly earlier in the lifespan, significantly reduce the risk of dementia and late-life cognitive decline.[20,34] Among former athletes with a higher history of concussions, which is not modifiable later in life, understanding the indirect pathway between cardiovascular health and neurological outcomes is especially important. The findings of the present study build upon prior research demonstrating an association between a greater history of concussion and an increased prevalence of CVD risk factors in former athletes under the age of 50 years.[17] These results have important clinical implications, highlighting the potential value of early screening, diagnosis, and intervention for individuals with a history of repeated concussion. Such strategies may mitigate the risk of dementia and other adverse neurological outcomes in later adulthood.

Early screening and detection represent essential components of effective intervention strategies for CVD risk factor management. Early identification and management of hypertension, hyperlipidemia, sleep apnea, and diabetes are associated with reduced risk of dementia and related cognitive decline.[33,35] For example, an earlier age of OSA diagnosis is linked to an increased long-term risk of developing dementia.[28] Importantly, early treatment of OSA, such as continuous positive airway pressure (CPAP), has been shown to reduce the overall risk of dementia.[35] In a cohort of older former football players, a diagnosis of OSA has also been associated with increased psychiatric and cognitive symptoms as well as increased tau positron emission tomography uptake, independent of age, race, BMI, and *APOE ε4* carrier status.[29] Furthermore, similar to what would be expected in the general population, associations between OSA and adverse outcomes were mitigated with treatment (CPAP) within this cohort of former players.[29] Given the higher rates of CVD risk factors in younger athletes with a greater history of concussion observed in this study, early detection and management of CVD risk factors, including hypertension, hyperlipidemia, and OSA, represent a modifiable pathway for future cognitive decline by which the risk of dementia might be significantly reduced.

Overlapping associations of concussion history, CVD risk factors and clinical function

Prior studies have reported significant associations of concussion history with higher symptom endorsement across multiple domains, including those in the current study (depression, cognition, anxiety, neurobehavioral symptoms), regardless of age.[9,21,22] Studies involving prospective longitudinal cohorts have demonstrated that psychiatric disorders are independent risk factors for subsequent CVD and associated risk factors.[36] Most pertinent to the current study, the onset of cardiovascular conditions was highest within 1 year of psychiatric diagnosis (younger age) and gradually declined over time when compared to matched siblings. Together, this suggests that the overlapping associations between psychiatric and CVD risk factors are stronger at younger ages, consistent with findings of the current study (along with concussion history). While the current study is cross-sectional in nature and the temporal association of symptom endorsement and cardiovascular risk factors cannot able to be determined, lower psychiatric and neurobehavioral function earlier in life associated with concussion history may lead to the subsequent onset of cardiovascular conditions consistent with the above-referenced longitudinal studies.

Mechanisms underlying concussion history and increased CVD burden

The mechanism underlying the association between cumulative concussion and a greater presence of cardiovascular risk factors remains unclear. The association is seen in nonsport mild traumatic brain injury (mTBI) and moderate to severe TBI, with a step-wise increase in strength with greater injury severity.[13] As noted above, higher concussion history is associated with greater levels of psychiatric and neurobehavioral symptom endorsement.[9,21,22] One possible explanation underlying the association includes postinjury behavioral changes in function associated with head injury, which may increase subsequent cardiovascular risk factor burden. In the general population, worse mental health is dynamically associated with unhealthy lifestyle behaviors, many of which increase the risk for poor cardiovascular health.[37-39] Other behavioral or inter-individual personality differences, such as predisposition for risk-taking behaviors, are associated with both head injury and cardiovascular risk, and may underlie the association between the two factors.[40,41]

Increased cardiovascular risk factor burden associated with head injury may also occur through several candidate biological mechanisms as well. For example, reduced heart rate variability is associated with new-

onset cardiovascular risk factors such as hypertension and hypercholesterolemia.[42,43] Decreased heart rate variability has been observed acutely following sport-related concussion, with mixed reports of chronic alterations in heart rate variability beyond clinical recovery.[44,45] Alterations in cardiovascular function following sport-related concussion have been reported by a select prospective study of collegiate football players. Specifically, as compared to preinjury baseline, increased pulse wave velocity and blood pressure were observed from 1 to 2 seasons longitudinally following sport-related concussion.[46] A cumulative effect of pro-inflammatory cytokine and chemokine elevations following acute concussion can contribute to vascular remodeling and disease as well.[47,48] Altogether, there is growing support for alterations in the cardiovascular system following sport-related concussion. Chronic subclinical alterations in autonomic nervous system function, pro-inflammatory cytokines, or other potential processes following clinical recovery of concussion (particularly recurrent injury) for years may be 1 pathway by which greater history of concussion can lead to higher cardiovascular risk factor rates/earlier onset in the younger age group.

Limitations

The self-reported nature of medical history (cardiovascular risk factors and concussion) is a limitation of the study. Although items inquired about "clinician diagnosed" disorders/diseases for cardiovascular risk factors, objectively measured function is the gold standard. Additionally, a prior study involving a subset of the NFL-LONG cohort with longitudinal data over 19 years (assessed at 9-year intervals) showed that the stability of self-reported concussion history reporting was fair to moderate, and that worse consistency (ie, increased number reported) was observed among those with higher levels of symptom endorsement.[49] This has the potential to bias or artificially inflate the association between cardiovascular risk factors and clinical outcomes, given, their mutual association with concussion history in the younger age group. Potential survival bias with enrollment could influence cardiovascular health outcomes in the older age group. However, an overall effect of age on cardiovascular risk factor prevalence was observed. Additionally, a wide distribution of concussion and cardiovascular risk factor presence was observed across both age groups, suggesting associations would still be detectable. The cross-sectional nature of the study design prevents conclusions surrounding the temporal nature of the association between concussion exposure, age, cardiovascular risk factors, and clinical function. True longitudinal studies are needed to determine the trajectories of cardiovascular risk among those with varying concussion histories.

CONCLUSION

Higher CVD risk factor disease burden is associated with a greater history of prior concussion in former athletes at younger, but not older ages. Furthermore, the presence of CVD risk factors is disproportionately associated with worse function across multiple domains in younger, as compared to older, former athletes. Given the higher rates of diagnoses at earlier ages, findings provide support for cardiovascular health as representing 1 pathway by which recurrent concussion may lead to adverse long-term outcomes in former contact sport athletes. A prevention approach, including early screening and management of cardiovascular risk factors in those with a greater history of concussion, should be considered in clinical practice.

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